USC University of Southern California

A Novel and Practical Approach to Generate All-Weather 30-Meter Land Surface Temperature Data

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authored with Lu Zhang and Siqin Wang

August 4th, 2025

IGARSS 2025, Brisbane

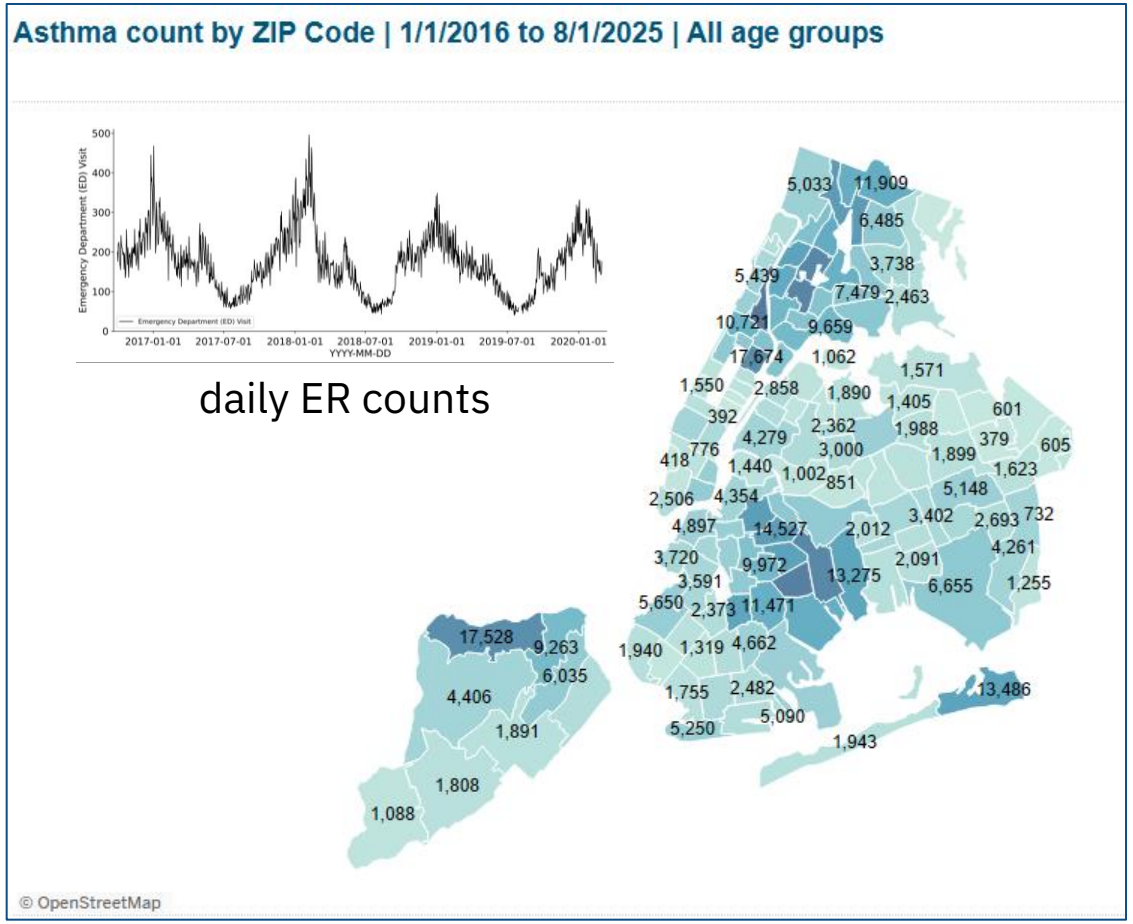
IGARSS Brisbane 2025

Background

Why we need daily temperature data at fine scale?

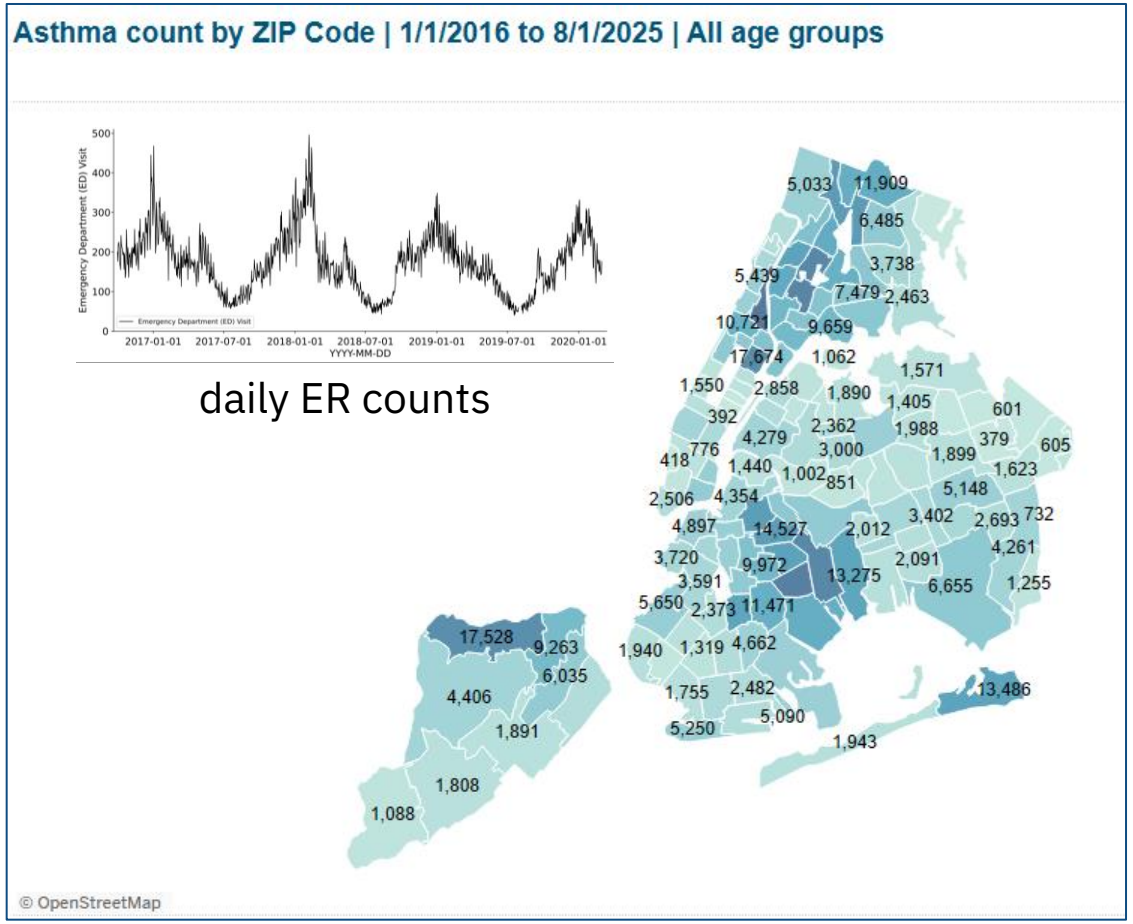
Daily, 30 meters

We need daily temperature data at fine scale



New York City. Daily number of people going to emergency room, by ZIP code

We need daily temperature data at fine scale



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For proper time series analysis, we need daily temperature data

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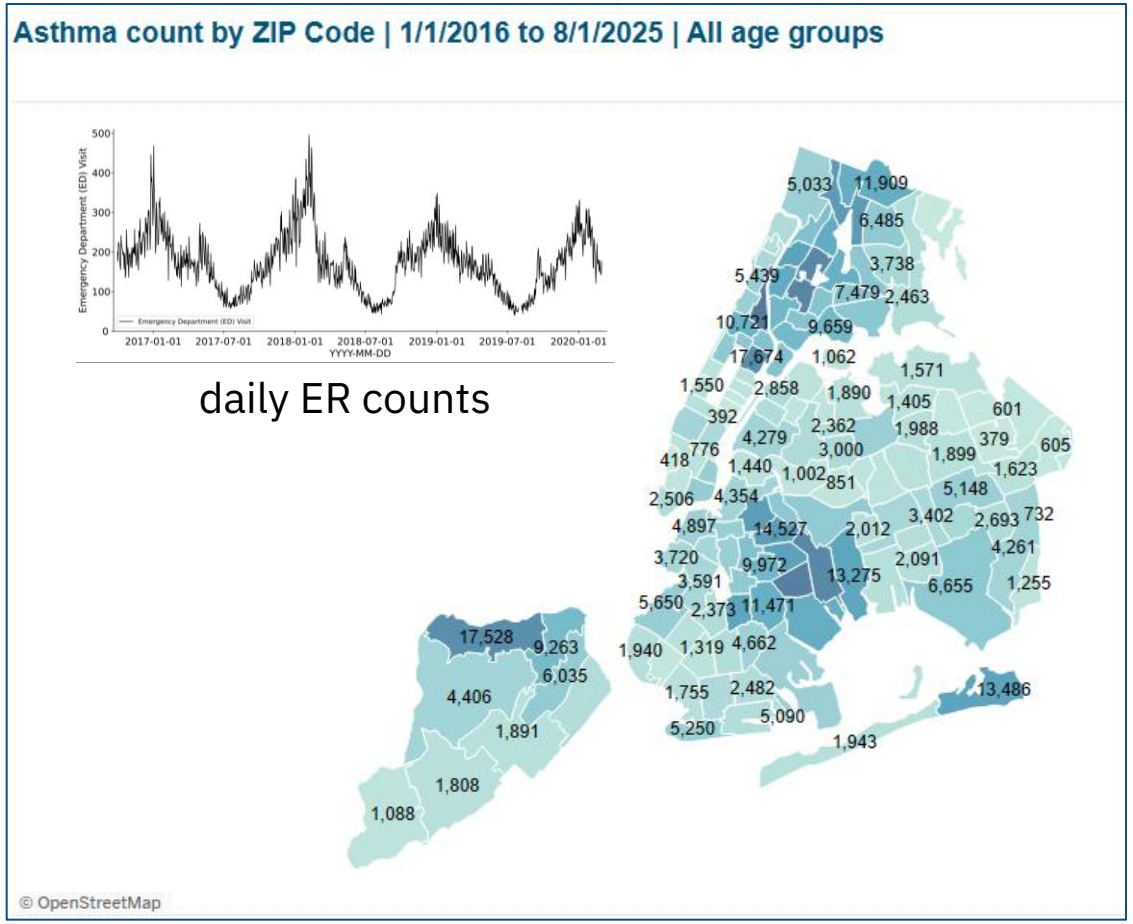

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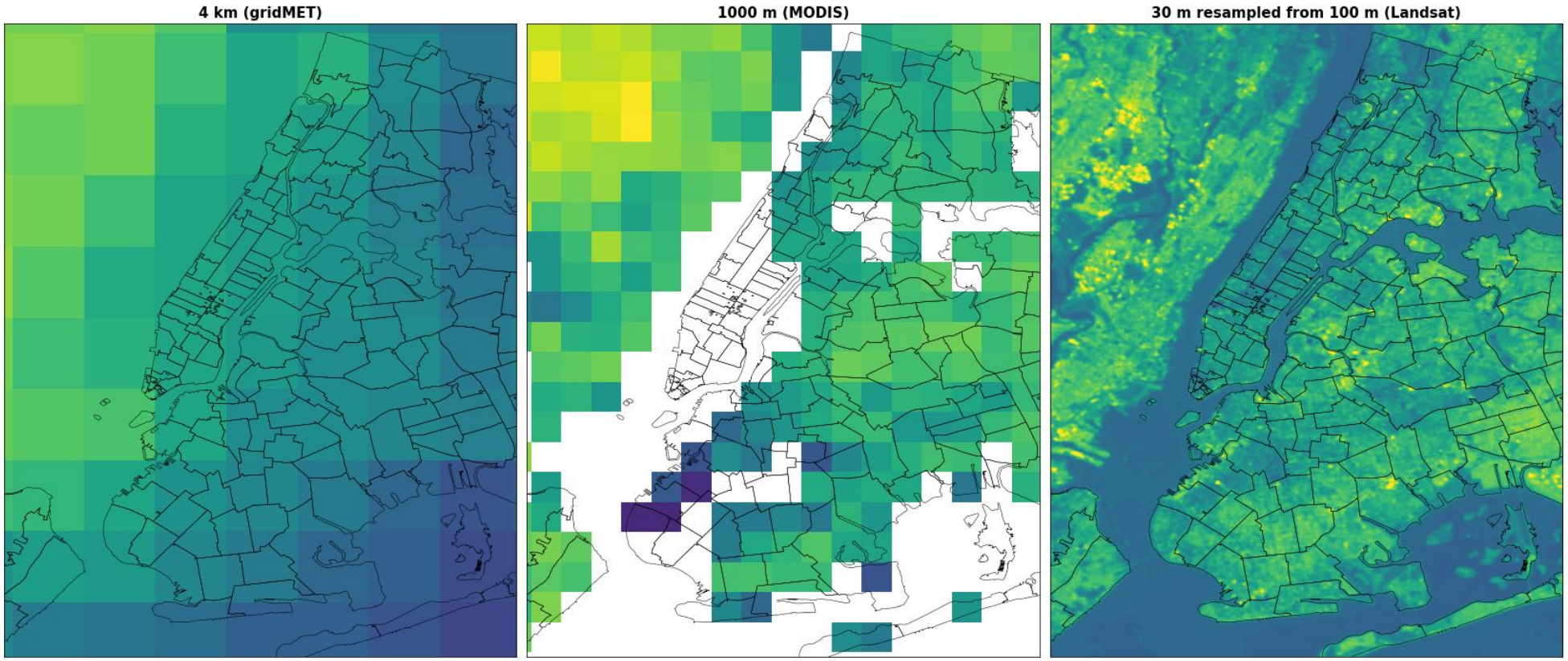
Temperature data requirements

- **Daily (temporal)**
- **ZIP code level (spatial)**

We don’t have good enough temperature data

We need daily temperature data at fine scale

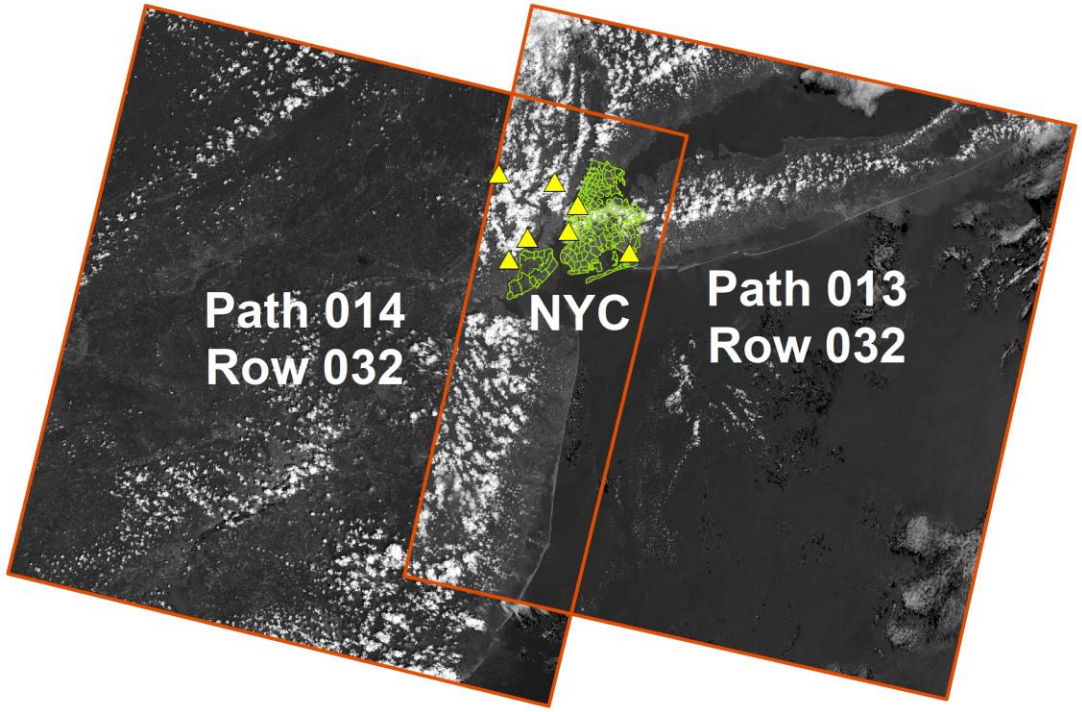


current best available data used
in climate and health research

[Garcia et al. 23]

Landsat LST:
high resolution + long-term record

Challenge to have daily Landsat temperature data



Clouds

not simply gap filling

need to consider **surface energy balance**

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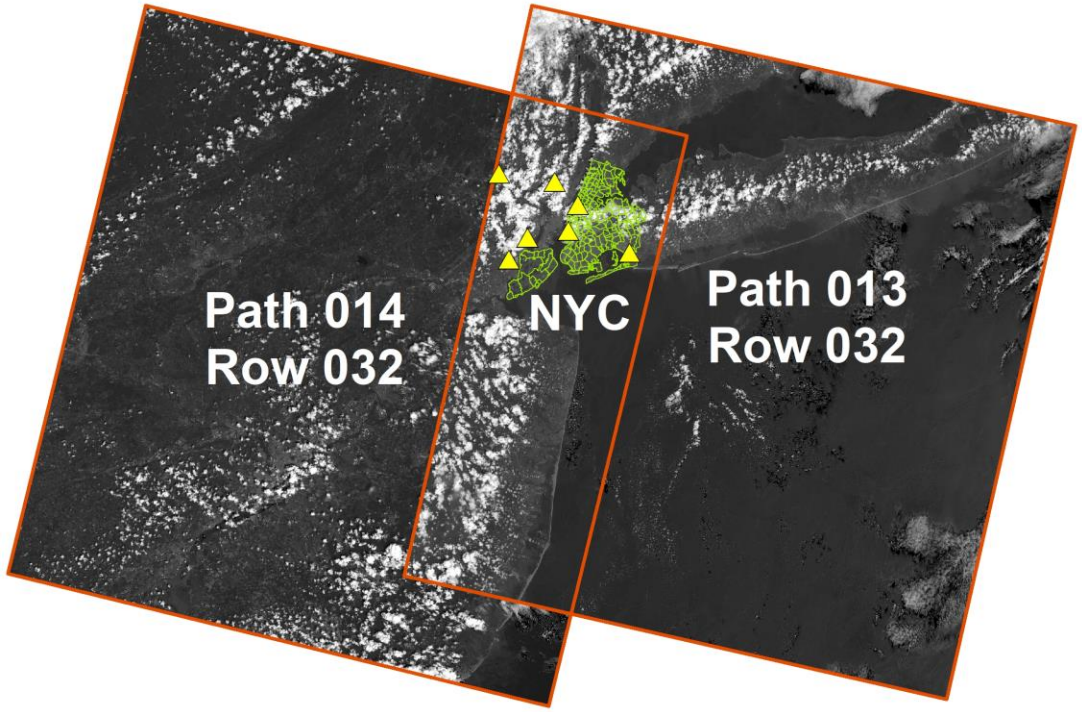

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1 scene / 16 days

2 scenes / 16 days (cross-track)

4 scenes / 16 days (cross-track, dual satellites)

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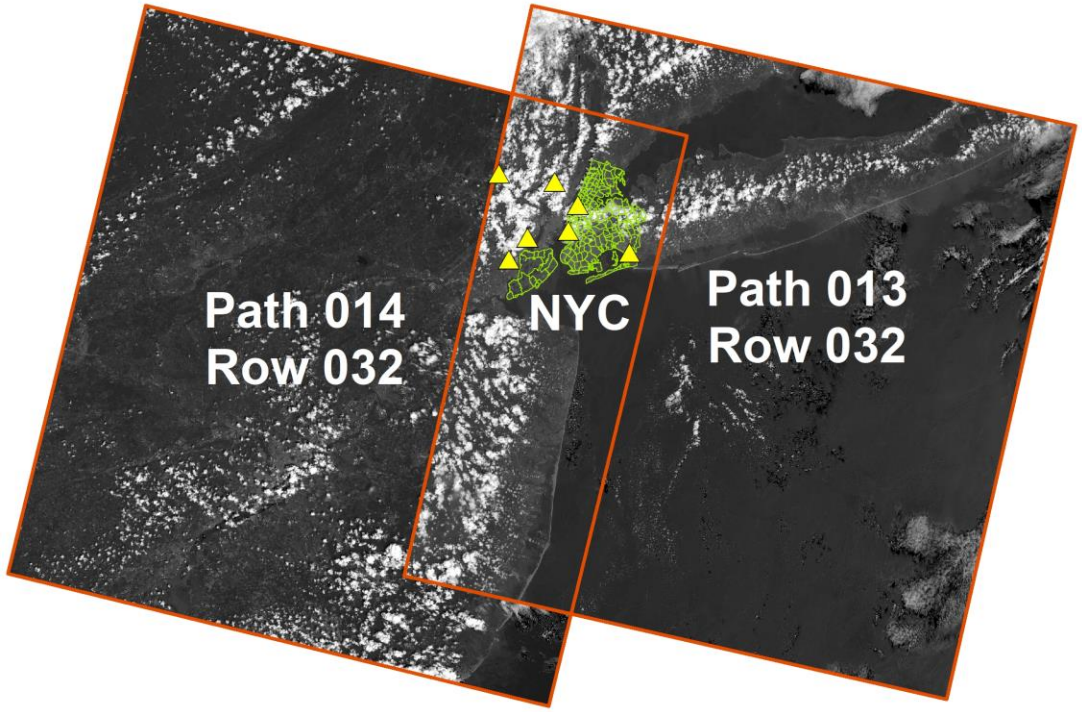

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No *in situ* LST site

validation under cloud-covered conditions?

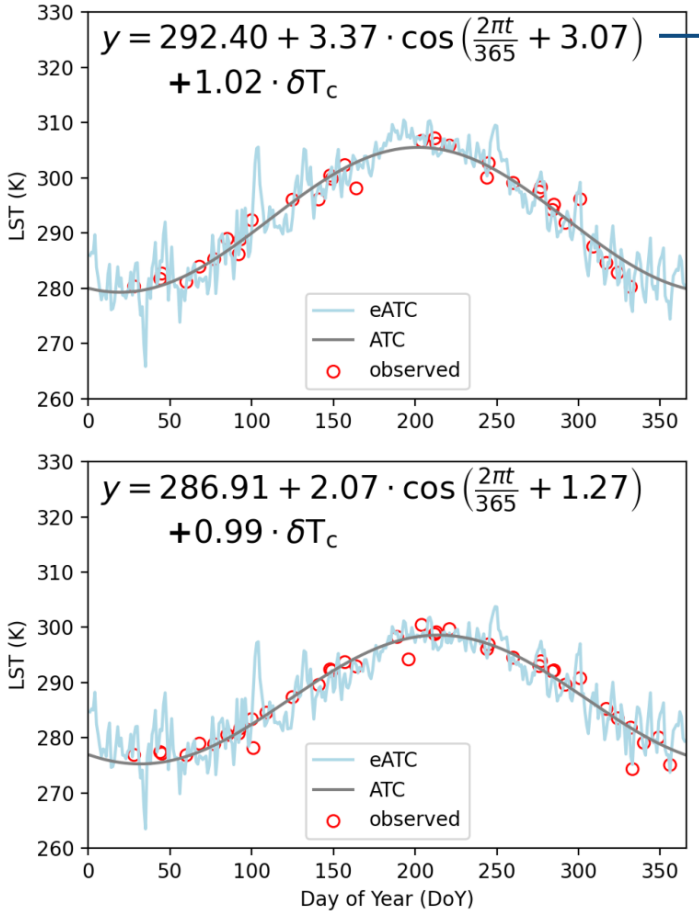
overfit in rural sites, performance in urban areas?

[Jia, Aolin et al. GRSM 2024]

Solution

Method

eATC: enhanced annual temperature cycle

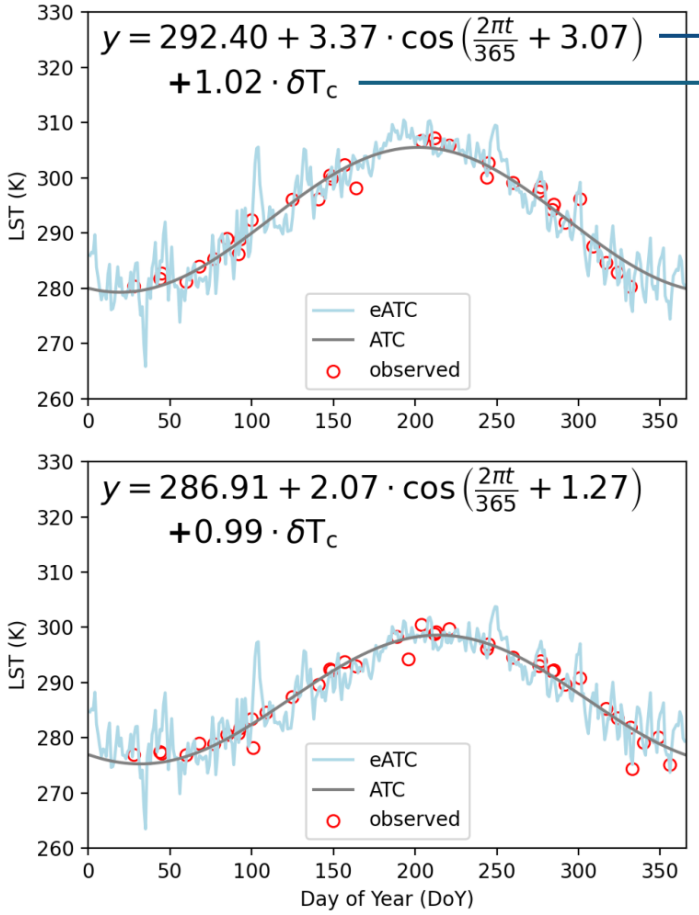


ATC

ATC

$$\mathcal{T}_s^R(d) = \boxed{C_s + A_s \cos\left(2\pi/365\,(d - \phi_s)\right)} + b_s \mathcal{T}^c(d)$$

eATC: enhanced annual temperature cycle



ATC
enhanced with ERA5

ERA5: hourly 0.1° reanalysis skin temperature via Earth system modeling, considering surface energy balance

Generated by European Centre for Medium-Range Weather Forecasts (ECMWF)

ATC ERA5

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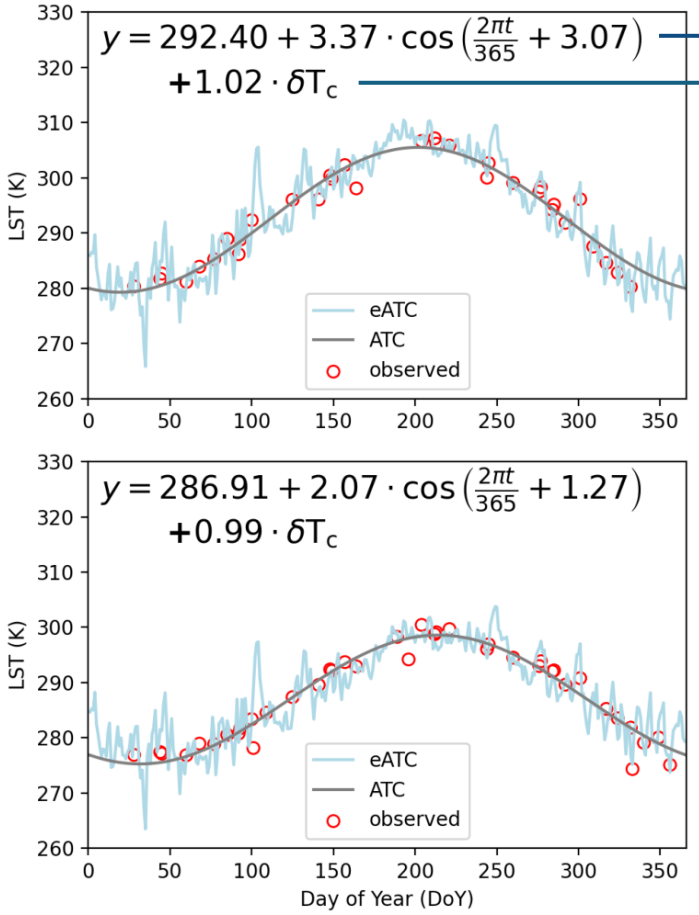

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Why It Works

1. Pixel-wise modeling: 4 parameters per pixel
   - Captures temporal fluctuations at the pixel level
   - Preserves spatial heterogeneity across the domain

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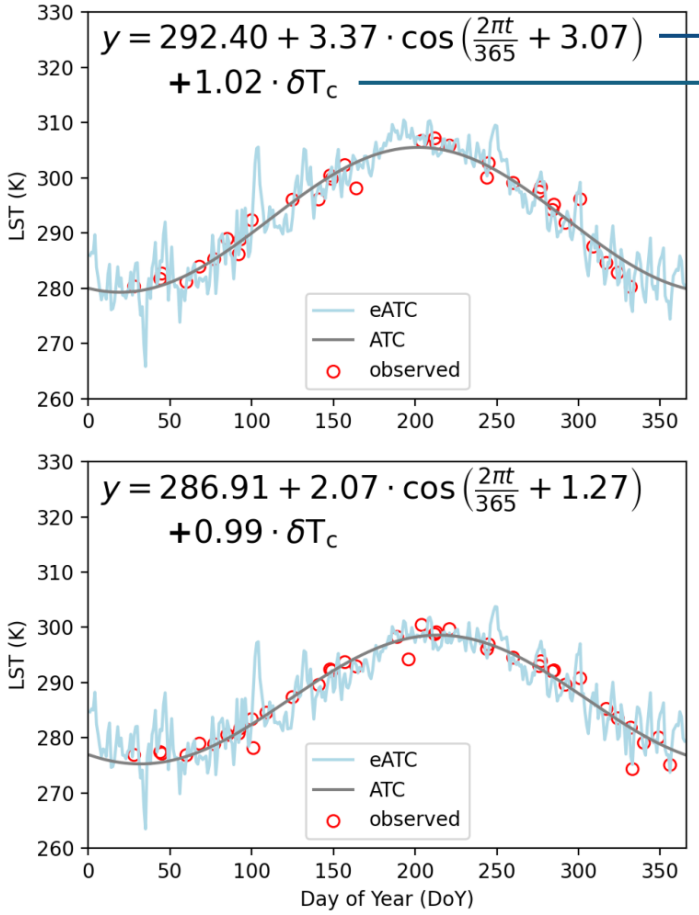

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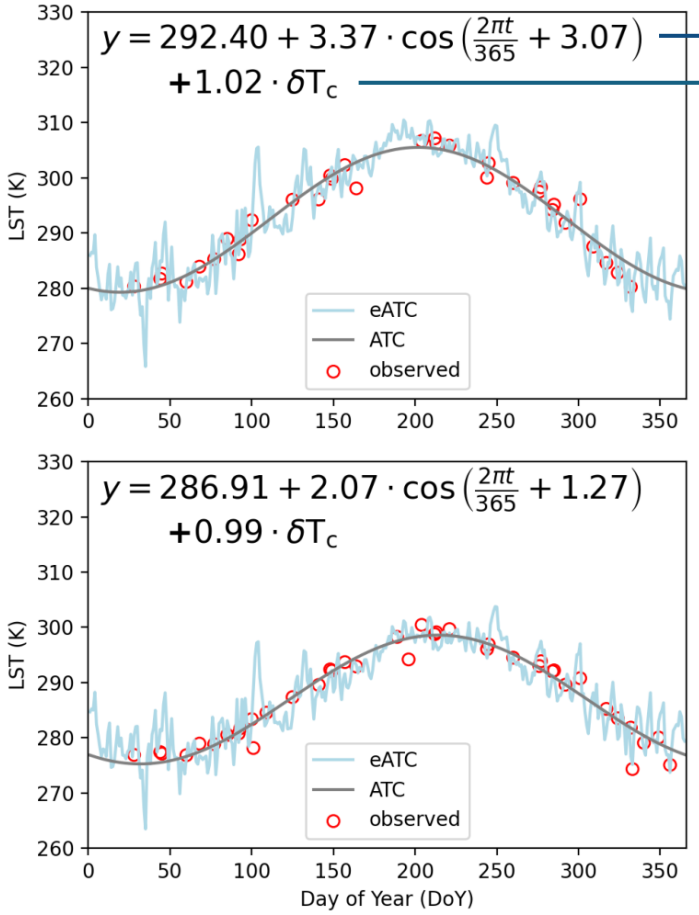

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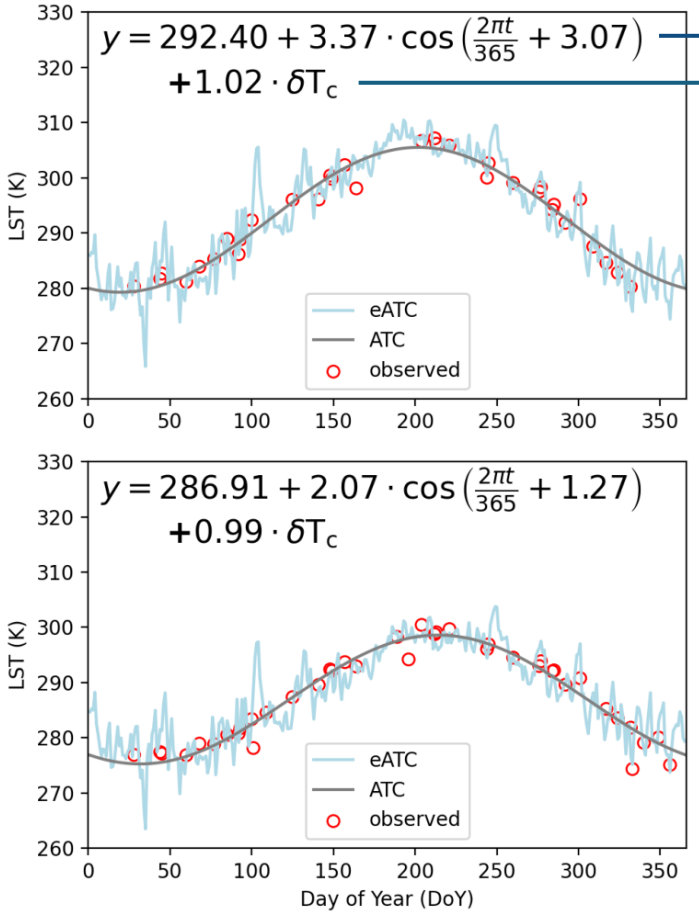

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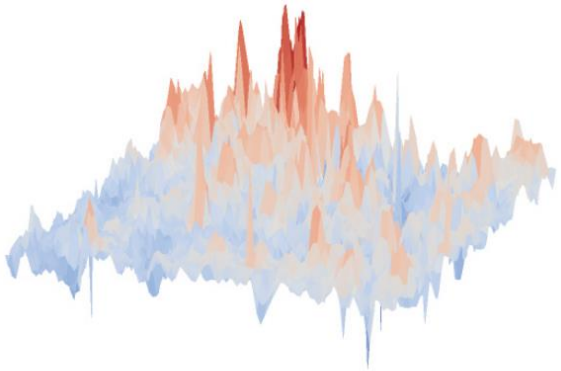
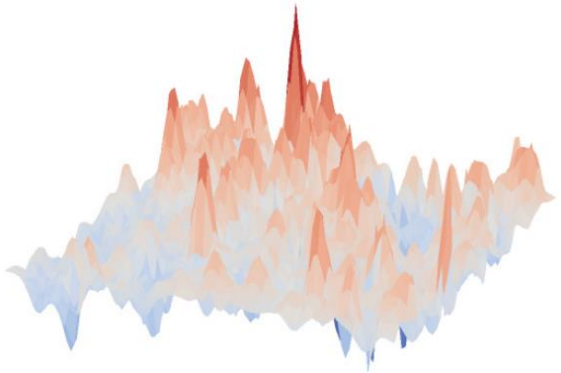
**Matrix-based GPU-accelerated computing with PyTorch

[for days with valid observations only]

Gaussian processes
Remaining residual correction



Spatial Residuals



Gaussian Processes Reconstruction

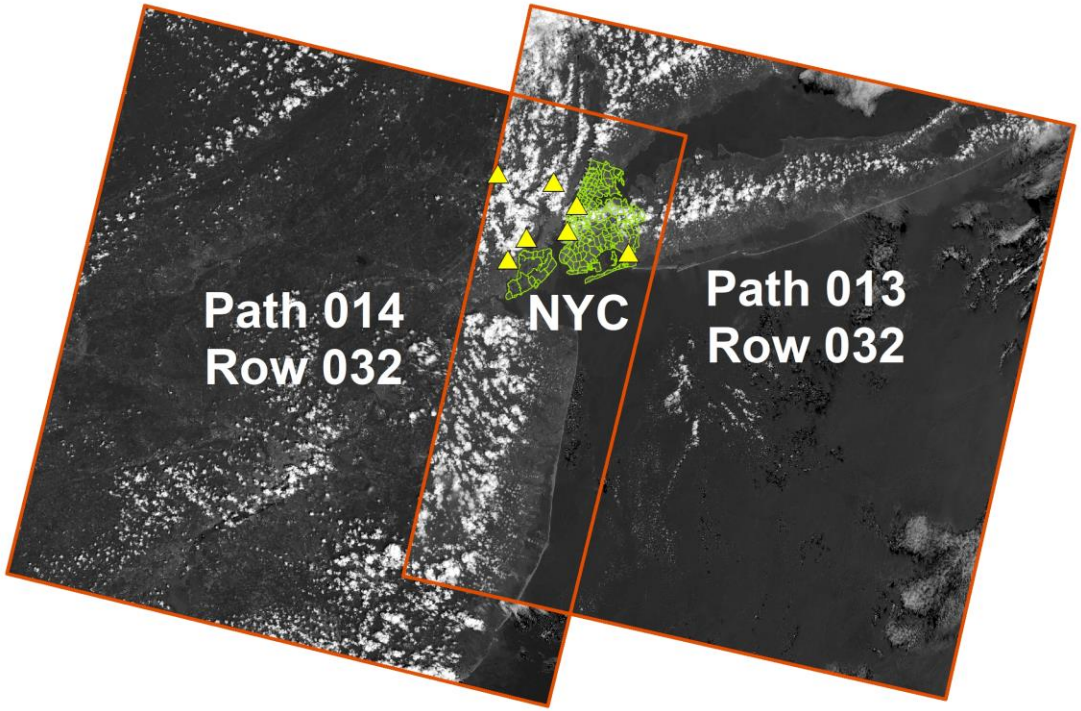
ATC ERA5

$$\mathcal{T}_s^R(d) = C_s + A_s \cos\left(2\pi/365\,(d - \phi_s)\right) + b_s \mathcal{T}^c(d) + \mathcal{GP}_d(s, \mathbf{s}_d)$$

residual surfaces are related to land cover

mean spectral bands as features in Gaussian processes

Data



91 Landsat scenes from 2023

(dual paths, dual satellites)

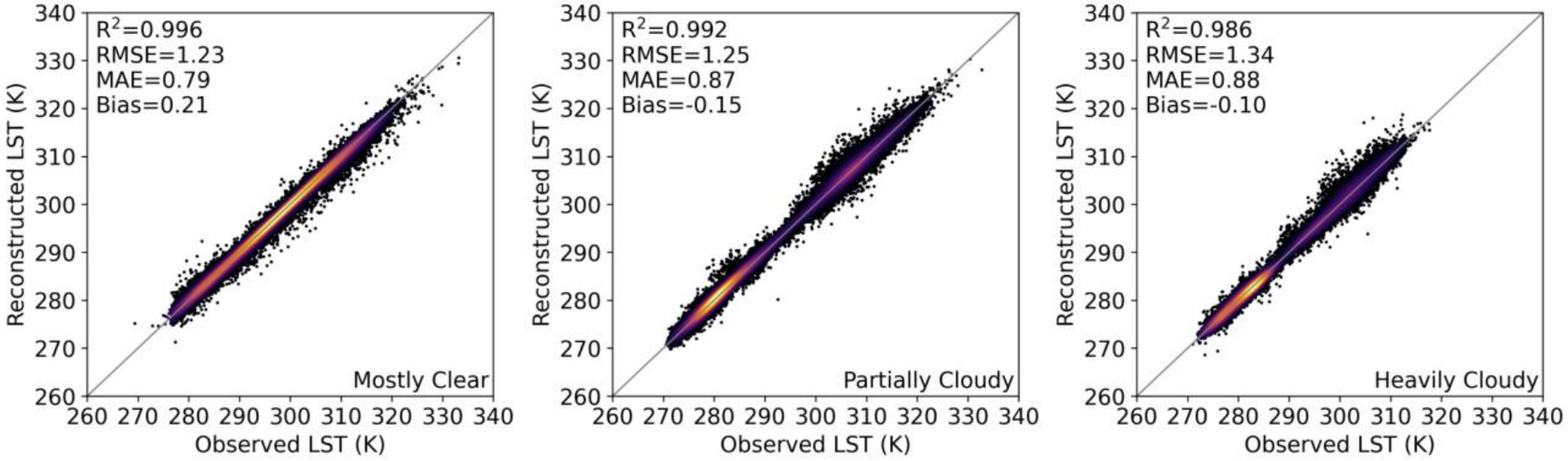
in situ air temperature data

(from 7 weather stations)

mean surface reflectance 2022-2024

Experiment #1

20% hold-out for validation, **all-weather**

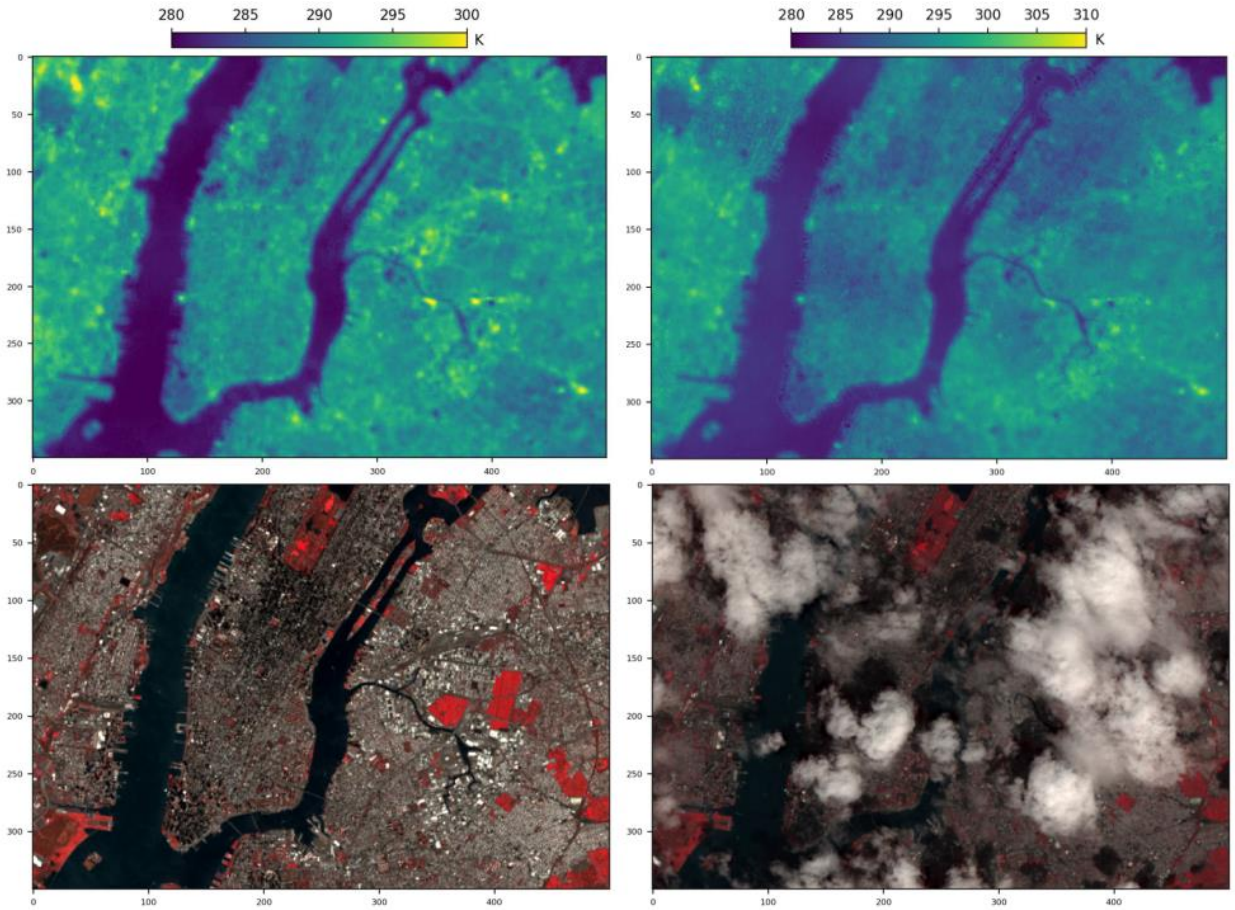


possible critics:

1. clear-sky reconstruction only
   → not true, this is all-weather reconstruction
2. training and validation pixels adjacent to each other
   → could have an impact due to Gaussian processes
   → would not affect days without observations

Experiment #1

Reconstruction results

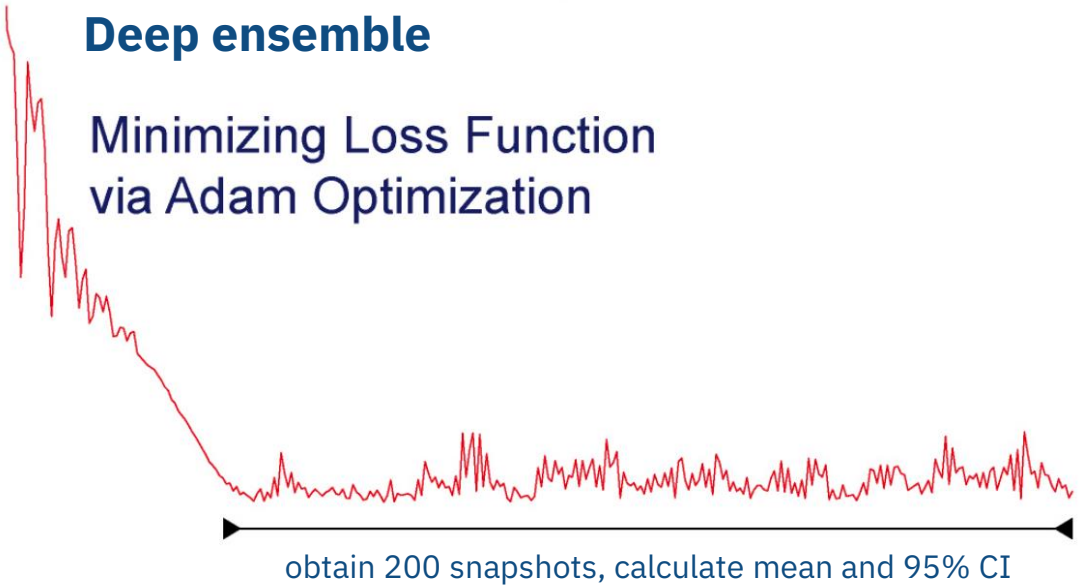


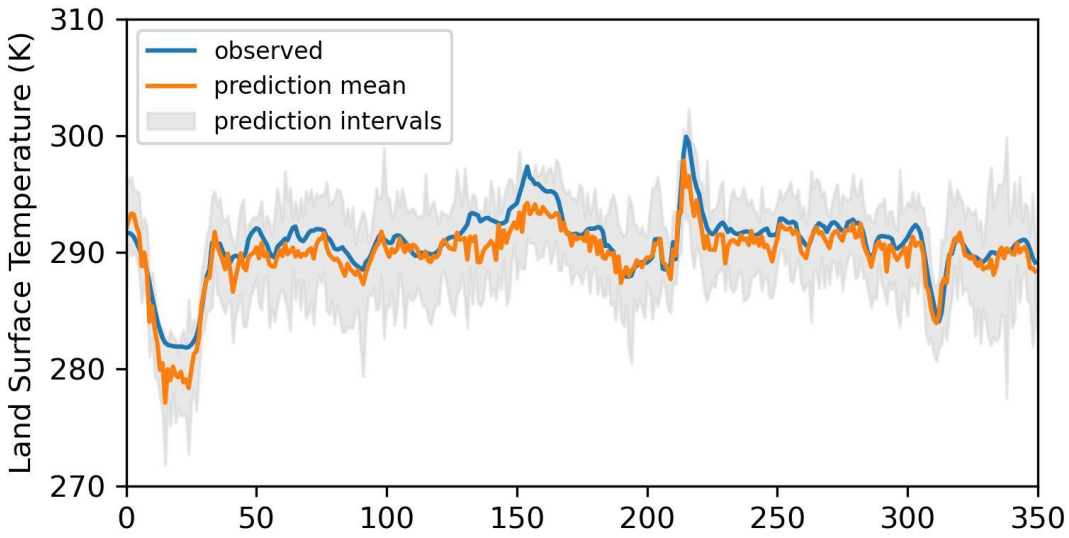
Uncertainty Quantification

Deep ensemble: 1st component of uncertainty

Gaussian processes: 2nd component of uncertainty

Combined is total uncertainty quantification [law of total variance]





Experiment #2

Indirect validation via estimating air temperature

Strategy

1. simple linear regression to transform LST to air temperature
2. separately for valid-observed and reconstructed pixels
3. if comparable with each other, reconstruction is successful

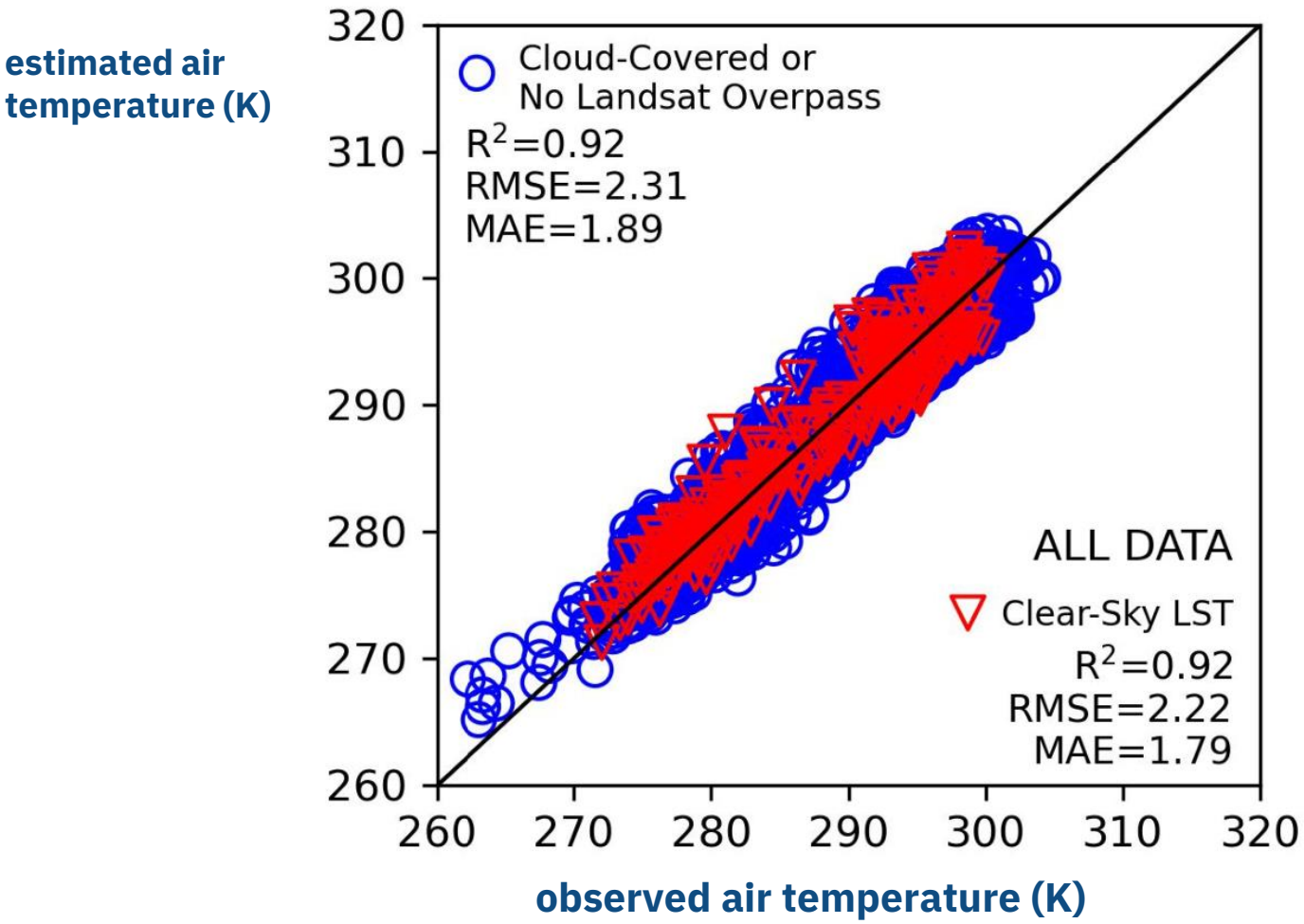
considering NDVI, elevation, solar radiation factor, solar zenith angle

$$\mathcal{T}_a = c + \mathcal{T}_{lst} \times \alpha_1 + V_{ndvi} \times \alpha_2 + V_{elv} \times \alpha_3 + V_{srf} \times \alpha_4 + V_{sza} \times \alpha_5$$

Benali et al. RSE 2012, Yang, Yujia et al. TGRS 2024

Experiment #2

Indirect validation with *in situ* air temperature



valid-observed data:
RMSE = 2.22 K
reconstructed data:
RMSE = 2.31 K

Comparable with each other over 7 weather stations

Summary

1. Novel and practical approach:

    ATC + ERA5, with GPU accelerated computing

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One step forward in achieving daily temperature data at 30-m resolution

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Thank you for your attention!

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